

Lab 2(BCS551)-DDL Commands & Constraints

Objective:At the end of the assignments, participants will be able to understand basic DDL, Create table with constraints, Alter, Truncate, Drop and Rename

Creating Table with constraints:

Query1:

Create Table book(bookno INT,title VARCHAR(50),status CHAR(1),price FLOAT)

Output1:

```
SQL> Create Table book(bookno INT,title VARCHAR(50),status CHAR(1),price FLOAT)
```

Table BOOK created.

Elapsed: 00:00:00.016

Query2:

desc book

Output2:

```
SQL> desc book
```

Name	Null?	Type
BOOKNO		NUMBER(38)
TITLE		VARCHAR2(50)
STATUS		CHAR(1)
PRICE		FLOAT(126)

Query3:

```
CREATE TABLE book1(  
    bookno INT PRIMARY KEY,  
    title VARCHAR (50) NOT NULL,  
    status CHAR(1) DEFAULT 'Y',  
    price FLOAT NOT NULL CHECK(price>0)  
)
```

Output3:

```
SQL> CREATE TABLE book1(  
      bookno INT PRIMARY KEY,  
      title VARCHAR (50) NOT NULL,  
      status CHAR(1) DEFAULT 'Y',...
```

[Show more...](#)

Table B00K1 created.

Elapsed: 00:00:00.020

Query4:

Desc book1

Output4:

```
SQL> desc book1
```

Name	Null?	Type
BOOKNO	NOT NULL	NUMBER(38)
TITLE	NOT NULL	VARCHAR2(50)
STATUS		CHAR(1)
PRICE	NOT NULL	FLOAT(126)

Query5:

ALTER TABLE book1 ADD Author VARCHAR(50)

Output5:

```
SQL> ALTER TABLE book1 ADD Author VARCHAR(50)
```

Table B00K1 altered.

Elapsed: 00:00:00.038

Query6:

desc book1

Output6:

```
SQL> desc book1
```

Name	Null?	Type
BOOKNO	NOT NULL	NUMBER(38)
TITLE	NOT NULL	VARCHAR2(50)
STATUS		CHAR(1)
PRICE	NOT NULL	FLOAT(126)
AUTHOR		VARCHAR2(50)

Query7:

ALTER TABLE book1 DROP COLUMN Author

Output7:

```
SQL> ALTER TABLE book1 DROP COLUMN Author
```

```
Table BOOK1 altered.  
Elapsed: 00:00:00.034
```

Query8:

desc book1

Output8:

```
SQL> desc book1
```

Name	Null?	Type
BOOKNO	NOT NULL	NUMBER(38)
TITLE	NOT NULL	VARCHAR2(50)
STATUS		CHAR(1)
PRICE	NOT NULL	FLOAT(126)

Query9:

ALTER TABLE book1 MODIFY title VARCHAR(100)

Output9:

```
SQL> ALTER TABLE book1 MODIFY title VARCHAR(100)
```

Table B00K1 altered.

Elapsed: 00:00:00.017

Query10:

Desc book1

Output10:

```
SQL> desc book1
```

Name	Null?	Type
BOOKNO	NOT NULL	NUMBER(38)
TITLE	NOT NULL	VARCHAR2(100)
STATUS		CHAR(1)
PRICE	NOT NULL	FLOAT(126)

Query11:

ALTER TABLE book MODIFY title VARCHAR (100) NOT NULL

Output11:

```
SQL> ALTER TABLE book MODIFY title VARCHAR (100) NOT NULL
```

Table B00K altered.

Elapsed: 00:00:00.020

Query12:

desc book

Output12:

```
SQL> desc B00K
```

Name	Null?	Type
BOOKNO		NUMBER(38)
TITLE	NOT NULL	VARCHAR2(100)
STATUS		CHAR(1)
PRICE		FLOAT(126)

Query13:

ALTER TABLE book1 RENAME COLUMN title TO booktitle

Output13:

```
SQL> ALTER TABLE book1 RENAME COLUMN title TO booktitle
```

Table B00K1 altered.

Elapsed: 00:00:00.016

Query14:

Desc book1

Output14:

```
SQL> desc book1
```

Name	Null?	Type
BOOKNO	NOT NULL	NUMBER(38)
BOOKTITLE	NOT NULL	VARCHAR2(100)
STATUS		CHAR(1)
PRICE	NOT NULL	FLOAT(126)

Query15:

DROP TABLE book

Output15:

```
SQL> DROP TABLE book
```

Table B00K dropped.

Elapsed: 00:00:00.027

Query16:

Desc book

Output16:

```
SQL> desc B00K
```

```
ERROR:
ORA-04043: object B00K does not exist
Error at Line: 4 Column: 0
```

Query17:

truncate table book1

Output17:

```
SQL> truncate table book1
```

Table BOOK1 truncated.

Elapsed: 00:00:00.007

Query18:

```
CREATE TABLE department1(  
    dept_id CHAR (3) PRIMARY KEY ,  
    dept_name CHAR(5) UNIQUE NOT NULL,  
    dept_location VARCHAR(20) NOT NULL  
)
```

Output18:

```
SQL> CREATE TABLE department1(  
    dept_id CHAR (3) PRIMARY KEY ,  
    dept_name CHAR(5) UNIQUE NOT NULL,  
    dept_location VARCHAR(20) NOT NULL...  
Show more...
```

Table DEPARTMENT1 created.

Elapsed: 00:00:00.022

Query19:

desc DEPARTMENT1

Output19:

```
SQL> desc DEPARTMENT1
```

Name	Null?	Type
DEPT_ID	NOT NULL	CHAR(3)
DEPT_NAME	NOT NULL	CHAR(5)
DEPT_LOCATION	NOT NULL	VARCHAR2(20)

Query20:

```
CREATE TABLE course1(  
  course_id CHAR(6) PRIMARY KEY,  
  course_name VARCHAR (50) UNIQUE NOT NULL,  
  course_credit NUMBER (2) NOT NULL,  
  dept_id CHAR(3) NOT NULL REFERENCES department (dept_id)  
)
```

Output20:

```
SQL> CREATE TABLE course1(  
      course_id CHAR(6) PRIMARY KEY,  
      course_name VARCHAR (50) UNIQUE NOT NULL,  
      course_credit NUMBER (2) NOT NULL,...  
Show more...
```

Table COURSE1 created.

Elapsed: 00:00:00.021

Query21:

```
desc course1
```

Output21:

```
SQL> desc course1
```

Name	Null?	Type
COURSE_ID	NOT NULL	CHAR(6)
COURSE_NAME	NOT NULL	VARCHAR2(50)
COURSE_CREDIT	NOT NULL	NUMBER(2)
DEPT_ID	NOT NULL	CHAR(3)

Query22:

```
INSERT INTO department1 VALUES ('D01','CS','ARYABHATTA BLOCK');
```

Output22:

```
SQL> INSERT INTO department1 VALUES ('D01','CS','ARYABHATTA BLOCK')
```

1 row inserted.

Elapsed: 00:00:00.024